



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 4 • STANDARD 6.AT.4

Use Percent to Solve Problems

Lesson 4-5 • Teacher Copy

Name: _____

Date: _____

Class: _____

My Notes

Level 3 Enrichment



TODAY'S OBJECTIVES

Content Objective: I can solve problems involving discounts, markups, tax, and tips using percents.

Language Objective: I can explain my solution using the words discount, markup, tax, tip, and percent.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Discount

Markup

Tax

Tip

Percent



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

An amount taken off the original price is a .

2

An amount added to a store's cost to set a higher selling price is a

.

3

An extra percent added to a purchase that goes to the government is

.

4

Extra money added for good service, often a percent of the bill, is a

.

5

A number out of 100 is a .

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

In your table you found the discount, then the tax. Why do we calculate the tax on the price AFTER the discount instead of the original price?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Discount** — Money taken off the first price to make it cheaper.

Descuento — Dinero que se quita del precio original para que sea más barato.

- ☐ **Markup** — Money added to the cost to set the selling price.

Recargo (margen) — Dinero que se suma al costo para fijar el precio de venta.

- ☐ **Tax** — Extra money added to a price for the government.

Impuesto — Dinero extra que se suma a un precio para el gobierno.

- ☐ **Percent** — A way to compare a number to 100, shown with the % sign.

Porcentaje — Una manera de comparar un número con 100, con el signo %.

- ☐ **Tip** — Extra money you give for good service.

Propina — Dinero extra que das por un buen servicio.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

I find the discount first because ____.

Primero hallo el descuento porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: A discount lowers the price, while tax, tip, and markup raise it.

Evidence: Students sort discount as lowering price and tax/tip/markup as raising it, and recognize all are a percent of a price.

Reasoning: This evidence connects the definition of percent to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **I find the discount first because ____.**
Primero hallo el descuento porque ____.
- **Then I take 5% tax on the ____ price, which is ____.**
Luego tomo 5% de impuesto sobre el precio ____, que es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
Mi afirmación es ____.
- **I know because ____.**
Lo sé porque ____.
- **So ____.**
Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.

- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.

- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.

- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.

- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Discount

2. **Notes 2:** Markup

3. **Notes 3:** Tax

4. **Notes 4:** Tip

5. **Notes 5:** Percent

6. **Watch for this mistake:** *A common mistake in Use Percent to Solve Problems is combining the discount and tax into one single percent change instead of applying them one after another. For a \$40 item with a 25% discount and 8% sales tax, a student might compute $40 \times (1 - 0.25 + 0.08) = 40 \times 0.83 = \33.20 , treating the tax as if it applies to the original price. But tax is charged on the discounted price, not the original price: first take 25% off \$40 to get $40 \times 0.75 = \$30$, then add 8% tax on that \$30: $30 \times 1.08 = \$32.40$. The correct final price is \$32.40, not \$33.20 — always compute the discount first, then apply tax to the new lower price.*

7. **Try It 1:** A. \$27 — Tax: $0.08 \times 25 = 2$. Total: $25 + 2 = \$27$.

8. **Try It 2:** A. \$6 — Tip: $0.15 \times 40 = \$6$.